JUNIOR DEOGRACIAS MAMBO MOTSOU

FPGA & SOC ENGINEER

VHDL
Vivado
Vitis

CONTACT

+33 6 01 16 86 06
motsoujunior17@gmail.com
teo1747.github.io/Portfolio
github.com/teo1747
linkedin.com/in/teo1747
France
Available now

LANGUAGES

English
native (anglophone), TOEIC C1 (2024)

French
fluent, professional use (B2)

TECHNICAL SKILLS

HDL & design

VHDL (synthesisable RTL, VHDL-2008 testbenches)
Verilog (basic) **FSMs**
pipelined datapaths **line buffers**
bit-growth management
AXI handshakes with back-pressure
bidirectional I/O

AMD/Xilinx flow

Vivado: IP integrator block design
IP packaging **synthesis**
implementation **XDC**
timing and utilisation reports
bitstream
ILA and Hardware Manager
Vitis HLS
Vitis bare-metal applications
TCL scripting of the flow

Devices & boards

Zynq-7000 (ZedBoard, xc7z020)
Artix-7 (Nexys 4 DDR, xc7a100t)

Interfaces & IP

AXI4-Lite **AXI4-Stream**
AXI DMA **Zynq PS7** (GP/HP ports)
Video Timing Controller
Clocking Wizard **FIFO Generator**
VGA **I²C** **UART** **SPI**
ZBT SRAM

Verification

Self-checking testbenches
bit-exact C reference models
random stall injection
GHDL **XSim** **ModelSim**
CI simulation on GitHub Actions

Embedded software

C (bare-metal, Xilinx standalone drivers, CMSIS)
FreeRTOS **Zephyr** **Python** **TCL**

Debug

ILA **JTAG / Hardware Manager**
J-Link **GDB** **serial consoles**

Embedded systems engineer (Polytech Dijon, 2026) focused on digital design for FPGAs and Zynq SoCs: RTL in VHDL, AXI4-Lite and AXI4-Stream interfaces, Vivado IP integration and block design, and the bare-metal software that drives the hardware from Vitis. I verify RTL against bit-exact C models with self-checking testbenches. A year of automotive firmware bring-up at FORVIA gives me the system view.

Professional experience

Sep. 2025 – Sep. 2026

FORVIA – Faurecia Interior Systems | Embedded Systems Engineer — apprentice (contrat de professionnalisation)

Surface Activation department, France

Firmware bring-up: LIN-controlled heating-element controller (TCU)

- Developed C firmware on the Infineon TLD4020-3ET (Arm Cortex-M23 with integrated LIN): LIN ON/OFF command handling, heater PWM through the LDRV peripheral, NTC-based temperature regulation, fault protection and a state-machine architecture.
- Diagnosed early bring-up blockers: traced a HardFault in device initialisation to a silicon-revision mismatch on the evaluation board (register absent on older silicon), and a lost debug connection to boot-mode jumper and USB power conditions.
- The NTC divider is supplied from the switched heater output, so the ADC node is a 100 Hz square wave during PWM. Synchronised sampling to the PWM on-time (ISR at mid-pulse, 16-sample averaging), which brought measurement scatter within the regulation requirement.

TLD4020-3STD_EVAL board **IAR Embedded Workbench**
Infineon MCU Configuration Wizard **on-board J-Link**
printf over LIN (MobaXterm, 19 200 baud 8N1)

MeLiBu LED driver port: Melexis MLX81116 to MLX81143

- Ported an existing MeLiBu 1.x driver on RP2040 to the MLX81143: extended-mode sub-address encoding (14 LEDs per node), linear 15-bit PWM encoding confirmed against the datasheet and on the hardware, for a 22×7 panel (154 RGB LEDs).
- Fixed bus-timing faults: break generation with busy-wait while interrupts are masked, critical section narrowed to the break to stop UART RX byte loss, TX-FIFO drain to prevent back-to-back frame collisions; corrected the bus return pin.
- Isolated the panel's LED wiring scramble in the hardware-mapping layer, separate from the input convention; used a PIO-based receiver to confirm slave acknowledgements.

RP2040 **Raspberry Pi Pico SDK** **PIO state machines** **C**
MeLiBu 1.x

Interior ambient-lighting demonstrator (software owner)

- Extended the control software of an automotive interior demonstration buck from a few LED displays to nine STM32 UART gateways driving WS2812 LED strips across several lighting zones.
- Wrote a multithreaded Python/PyQt5 application: OpenCV decodes MP4 animations, maps logical LED IDs to frame pixels and distributes each frame to one worker thread per gateway at 921 600 baud, with a barrier to keep all nine gateways in step; custom serial frames (ASCII with delta compression).
- Built an Android tablet app (Kotlin, Jetpack Compose) that controls zones, colours and brightness presets over Bluetooth serial using a JSON command protocol with explicit error codes; added a keyboard fallback mode.
- Adapted the STM32 gateway firmware with minimal changes (ID-offset filtering when two gateways share a UART line).

Python 3 **PyQt5** **OpenCV** **Kotlin/Jetpack Compose** **STM32** (C)
WS2812 **UART** **Bluetooth SPP** **Git**

Apr. – Aug. 2025

KMUTT — MDT Lab | Embedded Systems Intern

Bangkok, Thailand

- Image-processing work for a precision-agriculture project: fruit classification by transfer learning on VGG16 (Keras/TensorFlow), 98.4 % accuracy on a 4 599-image test set (Fruits-360, 131 classes), plus an OpenCV contour/colour detection pipeline.
- FPGA ramp-up: designed an 8-bit microcontroller in VHDL (12-state fetch–decode–execute controller, ALU with NZVC flags, memory-mapped ROM/RAM/I/O) following the LaMeres reference architecture; simulated in Vivado 2024.2 / XSim.

Python **TensorFlow/Keras** **OpenCV** **VHDL** **Vivado 2024.2**

FPGA & SoC projects

PERSONAL

Personal projects, verified in simulation and emulation, not yet on physical

CERTIFICATION

- Cisco CCNA 1 (July 2024)

boards.

Oct. 2026

Image-filter accelerator for Zynq-7000 with AXI DMA

personal · *github.com/teo1747/zynq-image-filter-accelerator*

- Streaming 3×3 filter IP in VHDL (box, Gaussian, Sobel), one pixel per clock: line buffers, 4-stage pipeline, full AXI4-Stream back-pressure, TLAST/TUSER framing; AXI4-Lite control and status registers; packaged for Vivado IP integrator.
- Zynq PS7 + AXI DMA block design scripted in TCL through bitstream and XSA; bare-metal Vitis application streams frames from DDR, checks every pixel against the C model and times HW against SW.
- Bit-exact against a C reference in a self-checking GHDL testbench with random stalls on both streams (14 frames, run in CI). Board run on ZedBoard pending.

VHDL **C** **TCL** **Vivado** **Vitis** **GHDL** **Zynq-7000** (xc7z020)

Oct. 2026 –

EmbSIM — microcontroller and circuit simulator

personal · *github.com/EmbLinkOs/EmbSIM*

- Instruction-set simulators and peripheral models (Cortex-M, AVR; STM32F1/F4 GPIO, timers, USART, SPI, I²C, ADC, DMA) coupled to a nodal-analysis circuit solver; CI runs unmodified STM32Cube, Arduino and Pico SDK examples. Early stage.

POLYTECH DIJON

2025 – 2026

Real-time VGA video pipeline on ZedBoard

Polytech, pair · *github.com/teo1747/zedboard-vga-video-pipeline*

- 640 × 480 chain built in IP integrator: Clocking Wizard, Video Timing Controller, AXI4-Stream video IPs, a VHDL inverter and Vitis HLS overlay IPs. Sync signals verified on hardware with ILA; the final demo showed no image because the HLS IPs had no AXI-Lite master to start them, since reworked as free-running (board retest pending).

Zynq-7000 (xc7z020) **Vivado/Vitis HLS 2022.2** **ILA** **XDC constraints**

2025 – 2026

ZBT SRAM controller in VHDL

Polytech SoC course, pair · *github.com/teo1747/vhdl-zbt-sram-controller*

- FSM controller for a Micron 512K × 36 ZBT SRAM: single and 4-word burst read/write on a 36-bit bidirectional bus (Xilinx IOBUF); self-checking testbench against the vendor memory model.

VHDL **Vivado** **Zynq-7000 target** **Micron simulation model**

Nov. 2024 – Jan. 2025

3×3 image filter on FPGA

Polytech, solo · *github.com/teo1747/vhdl-2d-image-filter*

- Streaming averaging and Gaussian (1-2-1/2-4-2/1-2-1) filters on 8-bit pixels: line buffers on the FIFO Generator IP, pipelined adder tree with explicit bit growth (9, 10, then 11 bits); simulated end-to-end on the Lena image, outputs compared with a software reference.

VHDL **Vivado 2024.1** **Artix-7** (xc7a100t) **XSim** **MATLAB**

2024 – 2025

FPGA temperature monitor

Polytech, pair · *github.com/teo1747/nexys4ddr-temperature-monitor*

- Integrated I²C (400 kHz) and UART (115 200 baud) cores to read the on-board ADT7420 and display it on the 7-segment display; timing closed at 100 MHz (WNS +5.4 ns, 1 936 LUTs) on Nexys 4 DDR.

Education

2023 – Sep. 2026

Engineering degree (Diplôme d'ingénieur, Master's level) — Embedded Systems | Polytech Dijon

Dijon, France

Graduated. Final-year apprenticeship at FORVIA. Coursework: digital design, SoC design on Zynq (Vivado, Vitis, Vitis HLS), real-time systems.

2021 – 2023

Two-year preparatory programme — mathematics, physics, electronics | IUC

Cameroon